

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V) Dot
.Net Core and Progressive Web Application Practical

Practical – X

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a. Create a PWA with a Web App Manifest file including:

App name

Icons

Start URL

Display mode

Implement Push Notifications in a PWA.

Code :

Manifest.json

```
{
  "name": "My PWA Application",
  "short_name": "MyPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon2.png",
      "sizes": "128x128",
      "type": "image/png"
    },
    {
      "src": "icons/icon1.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

Index.html

```

13 <h1>My Progressive Web App T023</h1>
14
15 <button onclick="showNotification()">
16     Show Notification
17 </button>
18
19 <script src="app.js"></script>
20
21 </body>
22 </html>

```

App.js

```

if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('service-worker.js')
        .then(() => console.log('Service Worker Registered'));
}

0 references
function showNotification() {
    Notification.requestPermission().then(permission => {
        if (permission === 'granted') {
            navigator.serviceWorker.ready.then(registration => {
                registration.showNotification('PWA Notification', {
                    body: 'Push Notification Working Successfully!',
                    icon: 'icons/icon1.png'
                });
            });
        }
    });
}

```

Service-worker.js

```

self.addEventListener('install', event => {
    console.log('Service Worker Installed');
});
self.addEventListener('activate', event => {
    console.log('Service Worker Activated');
});

```

Program.cs

```

var builder = WebApplication.CreateBuilder(args);

var app = builder.Build();

app.UseDefaultFiles();

app.UseStaticFiles();

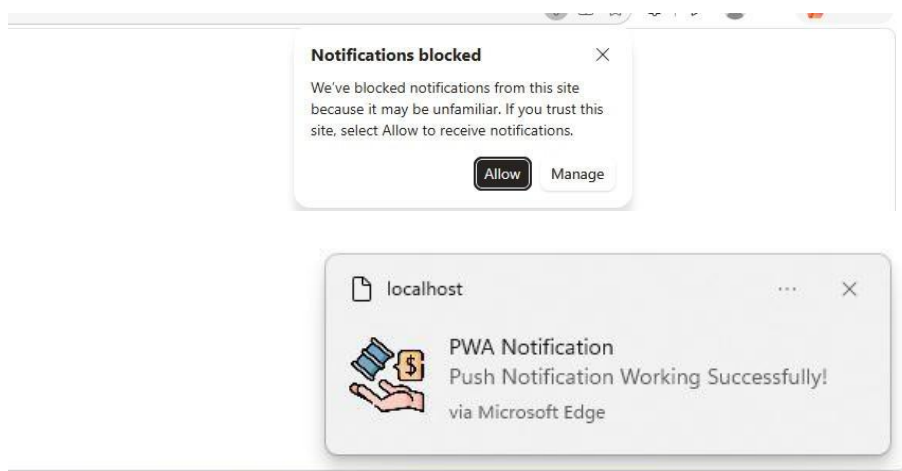
app.Run();

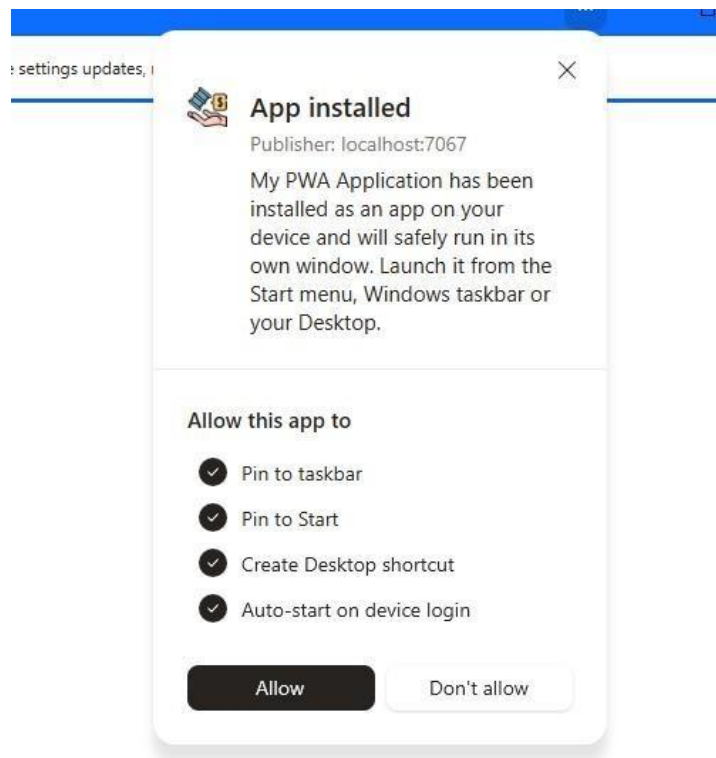
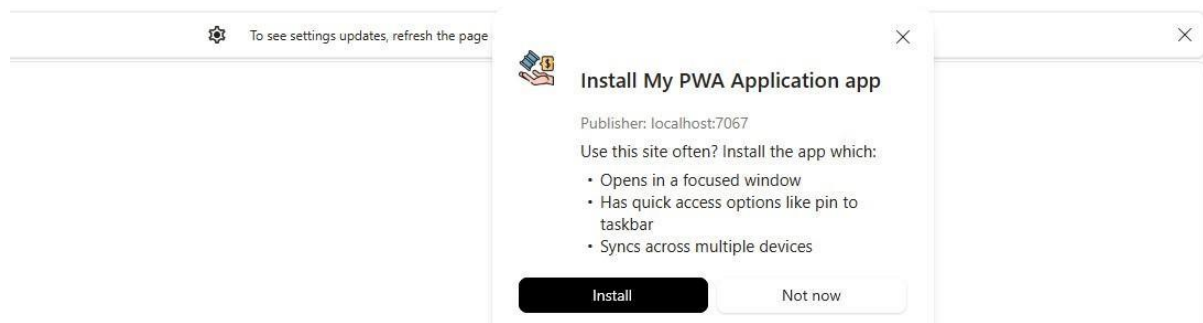
```

Output :

My Progressive Web App T023

Show Notification





b. Develop an E-Commerce PWA with product listing and offline cart functionality.

Code:

Program.cs

```
service-worker.js  app.js  index.html  manifest.json  Program.cs
ECommercePWA
1  var builder = WebApplication.CreateBuilder(args);
2  var app = builder.Build();
3
4  app.UseDefaultFiles();
5  app.UseStaticFiles();
6
7  app.Run();
```

Manifest.json:

```
service-worker.js  app.js  index.html  manifest.json*  Program.cs
Schema: https://www.schemastore.org/foxxx-manifest.json
1  {
2    "name": "E-Commerce PWA",
3    "short_name": "ShopPWA",
4    "start_url": "/index.html",
5    "display": "standalone",
6    "background_color": "#ffffff",
7    "theme_color": "#0d6efd",
8    "icons": [
9      {
10       "src": "icons/icon1.png",
11       "sizes": "128x128",
12       "type": "image/png"
13     },
14     {
15       "src": "icons/icon2.png",
16       "sizes": "512x512",
17       "type": "image/png"
18     }
19   ]
20 }
21
```

Index.html:

```
service-worker.js  app.js  index.html  manifest.json*  Program.cs
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>E-Commerce PWA</title>
5    <link rel="manifest" href="manifest.json">
6    <meta name="theme-color" content="#0d6efd">
7
8    <style>
9      body {
10        font-family: Arial;
11        margin: 20px;
12      }
13
14      .container {
15        display: flex;
16        gap: 30px;
17      }
18
19      .cart-section {
20        width: 30%;
21        border: 1px solid gray;
22        padding: 15px;
23        min-height: 300px;
24      }
25
26      .product-section {
27        width: 70%;
28      }
29
30      .products {
31        display: flex;
32        flex-wrap: wrap;
33        gap: 15px;
34      }
35
36      .product {
37        border: 1px solid gray;
38        padding: 10px;
39        width: 200px;
40      }
41    </style>
42  </head>
43  <body>
44    <div class="container">
45      <div class="cart-section">
46        <h3>Shopping Cart</h3>
47        <table>
48          <thead>
49            <tr>
50              <th>Product</th>
51              <th>Quantity</th>
52              <th>Total</th>
53            </tr>
54          </thead>
55          <tbody>
56            <tr>
57              <td>Product 1</td>
58              <td>1</td>
59              <td><span></span></td>
60            </tr>
61            <tr>
62              <td>Product 2</td>
63              <td>1</td>
64              <td><span></span></td>
65            </tr>
66          </tbody>
67        </table>
68      </div>
69      <div class="product-section">
70        <h3>Products</h3>
71        <div class="products">
72          <div class="product">
73            <img alt="Product 1 image" data-bbox="100 100 150 150"/>
74            <h4>Product 1</h4>
75            <p>Description of Product 1</p>
76            <p><span></span></p>
77          </div>
78          <div class="product">
79            <img alt="Product 2 image" data-bbox="100 100 150 150"/>
80            <h4>Product 2</h4>
81            <p>Description of Product 2</p>
82            <p><span></span></p>
83          </div>
84        </div>
85      </div>
86    </div>
87  </body>
88  </html>
```

App.js:

```
service-worker.js  app.js  index.html  manifest.json*  Program.cs
ECommercePWA JavaScript Content Files  displayCart
1  if ('serviceWorker' in navigator) {
2      navigator.serviceWorker.register('service-worker.js')
3          .then(() => console.log('Service Worker Registered'));
4  }
5
6  let cart = JSON.parse(localStorage.getItem('cart')) || [];
7  displayCart();
8
9  0 references
10 function addToCart(product) {
11     cart.push(product);
12     localStorage.setItem('cart', JSON.stringify(cart));
13     displayCart();
14 }
15
16 2 references
17 function displayCart() {
18     let cartList = document.getElementById('cartList');
19     cartList.innerHTML = '';
20
21     cart.forEach(item => {
22         let li = document.createElement('li');
23         li.textContent = item;
24         cartList.appendChild(li);
25     });
26 }
```

Service-worker.js

```
service-worker.js  app.js  index.html  manifest.json*  Program.cs
ECommercePWA JavaScript Content Files  <global>
1  const CACHE_NAME = 'ecommerce-cache-v1';
2
3  const urlsToCache = [
4      '',
5      '/index.html',
6      '/app.js',
7      '/manifest.json'
8  ];
9
10 self.addEventListener('install', event => {
11     event.waitUntil(
12         caches.open(CACHE_NAME)
13             .then(cache => {
14                 return cache.addAll(urlsToCache);
15             })
16     );
17 });
18
19 self.addEventListener('fetch', event => {
20     event.respondWith(
21         caches.match(event.request)
22             .then(response => {
23                 return response || fetch(event.request);
24             })
25     );
26 });
```

Output:

E-Commerce PWA

Cart Items

- Headphones
- Laptop
- Mobile
- Laptop

Products

Laptop

Price: ₹50000

Add to Cart

Mobile

Price: ₹20000

Add to Cart

Headphones

Price: ₹3000

Add to Cart

